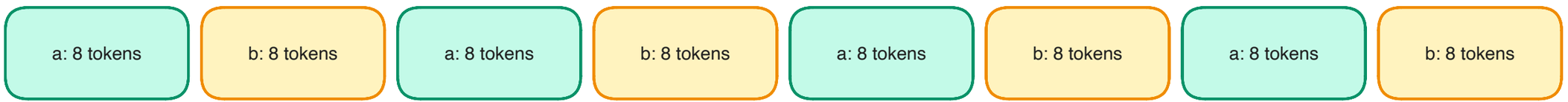
Why one bigger batch beats two replicas on one GPU

Each decode step reads all ~17 GiB of weights. What matters is how many calls share each read.

engine a's step engine b's step one engine, 16 calls

Two engines × 8 slots: they take turns on the GPU (conceptual schedule, not a measured trace)



every call waits for both:
14–17 ms per token

One engine × 16 slots: one read serves 16 calls



8 ms per token
~1,700 tokens/s (vs ~1,000)

Measured (H1): 24 users sustained instead of 16, 1.8× the goodput at 24 users (4.02 vs 2.20 calls/s).
Replicas still matter, for failover and across GPUs, but on one GPU batch first.